

2(a)(i)	$E = (\Delta)V / (\Delta)d$	C1
	$= 1340 / 1.4 \times 10^{-2}$	A1
	$= 9.6 \times 10^4 \text{ N C}^{-1}$	
2(a)(ii)	$F = Eq$ or $q(\Delta)V / (\Delta)d$	C1
	$q = 4.6 \times 10^{-14} / 9.6 \times 10^4$ or $4.6 \times 10^{-14} \times 1.4 \times 10^{-2} / 1340$	A1
	$= 4.8 \times 10^{-19} \text{ C}$	
	sign of charge: negative	B1
2(b)(i)	(adjacent field) <u>lines</u> have same separation (for both patterns)	B1
	(direction of lines changes from) downwards to upwards	B1

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Question	Answer	Marks
2(b)(ii)	$\text{resultant force} = 4.6 \times 10^{-14} + (9.6 \times 10^4 \times 4.8 \times 10^{-19})$ $= 4.6 \times 10^{-14} + 4.6 \times 10^{-14}$ $= 9.2 \times 10^{-14} \text{ N}$	A1
2(b)(iii)	$(a =) F/m \text{ or } 2W/m \text{ or } 2g$	B1
	$a = 9.2 \times 10^{-14} / (4.6 \times 10^{-14} / 9.81) = 20 \text{ (m s}^{-2}\text{)}$ or $a = 2 \times 9.81 = 20 \text{ (m s}^{-2}\text{)}$	A1
2(b)(iv)	$s = ut + \frac{1}{2}at^2$ $(1.4 \times 10^{-2} / 2) = \frac{1}{2} \times 20 \times t^2$	C1
	$t = 2.6 \times 10^{-2} \text{ s}$	A1
2(c)	line from $(0, 0.7 \times 10^{-2})$ to a non-zero point on the t -axis	M1
	magnitude of gradient of line increases	A1